Getting Started

Chapter 1

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User Manual Overview

Overview of this Publication

The *DURAPULSE* GS10 Drive User Manual describes the installation, configuration, and methods of operation of the *DURAPULSE* GS10 Series AC Drive. Throughout this manual, please note:

- *GS10 refers to GS11 and GS13 models only*

Who Should Read This Manual

This manual contains important information for those who will install, maintain, and/or operate any of the GS10 Series AC Drives.

Supplemental Publications

The National Electrical Manufacturers Association (NEMA) publishes many different documents that discuss standards for industrial control equipment. Global Engineering Documents handles the sale of NEMA documents. For more information, you can contact Global Engineering Documents at:

15 Inverness Way East
Englewood, CO 80112-5776
1-800-854-7179 (within the U.S.)
303-397-7956 (international)
www.global.ihs.com

Technical Support

By Telephone: 770-844-4200
(Mon.–Fri., 9:00 a.m.–6:00 p.m. E.T.)

On the Web: www.automationdirect.com

Our technical support group is glad to work with you in answering your questions. If you cannot find the solution to your particular application, or, if for any reason you need additional technical assistance, please call technical support at **770-844-4200**. We are available weekdays from 9:00 a.m. to 6:00 p.m. Eastern Time.

We also encourage you to visit our web site where you can find technical and non-technical information about our products and our company. Visit us at www.automationdirect.com.

Special Symbols

NOTE: When you see the "notepad" icon in the left-hand margin, the paragraph to its immediate right will be a special note.

WARNING: When you see the "exclamation mark" icon in the left-hand margin, the paragraph to its immediate right will be a warning. This information could prevent injury, loss of property, or even death (in extreme cases).

Purpose of AC Drives

AC drives are generally known by many different names: Adjustable Frequency Drives (AFD), Variable Frequency Drives (VFD), and Inverters. Drives are used primarily to vary the speed of three phase AC induction motors, and they also provide non-emergency start and stop control, acceleration and deceleration, and overload protection. By gradually accelerating the motor, drives can reduce the amount of motor startup inrush current.

AC drives function by converting incoming AC power to DC, which is then synthesized back into three phase output power. The voltage and frequency of this synthesized output power is directly varied by the drive, where the frequency determines the speed of the three phase AC induction motor.

Selecting the Proper Drive Rating

Determine Motor Full-Load Amperage (FLA)

Motor FLA is located on the nameplate of the motor.
NOTE: FLA of motors that have been rewound may be higher than stated.

Determine Motor Overload Requirements

Many applications experience temporary overload conditions due to starting requirements or impact loading. Most AC drives are designed to operate at 150% overload for 60 seconds. If the application requires an overload greater than 150% or longer than 60 seconds, the AC drive must be oversized.

NOTE: Applications that require replacement of existing motor starters with AC drives may require up to 600% overload.

Determine Application Type; Constant Torque or Variable Torque

This torque requirement has a direct effect on which drive to select. Variable Torque (VT) applications are generally easier to start; typically fans and pumps. Most other applications outside fans and pumps fall into the Constant Torque (CT) category (machine control, conveyors, etc.). If you are unsure of the application, assume Constant Torque. The specification, derating, and selection tables are generally segregated by Constant Torque and Variable Torque.

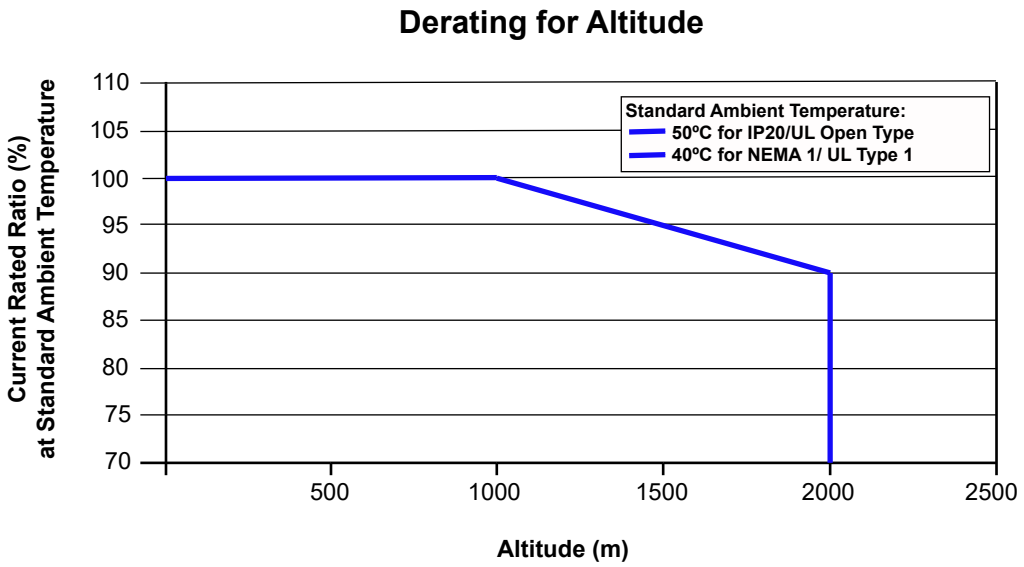
Installation Altitude

AC drives rely on air flow for cooling. As the altitude increases, the air becomes less dense, and this drop in air density decreases the cooling properties of the air. Therefore, the AC drive must be oversized to compensate for the decrease in cooling. Most AC drives are designed to operate at 100% capacity at altitudes up to 1000 meters.

NOTE: For use above 1000m, the AC drive must be derated as described below.

Derate Output Current Based on Altitude Above 1000 Meters

- *If the AC drive is installed at an altitude of 0~1000m, follow normal operation restrictions.*
- *If installed at an altitude of 1000~2000m, decrease 1% of the rated current or lower 0.5°C of temperature for every 100m increase in altitude.*
- *Maximum altitude for Corner Grounded is 2000m. If installation at an altitude higher than 2000m is required, please contact AutomationDirect.*



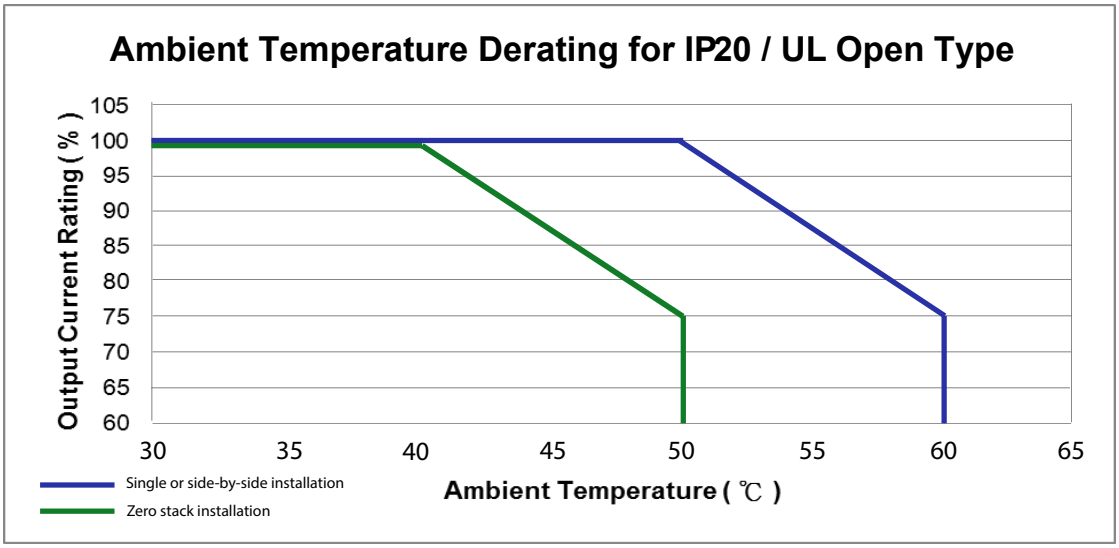
Determine Maximum Enclosure Internal Temperature

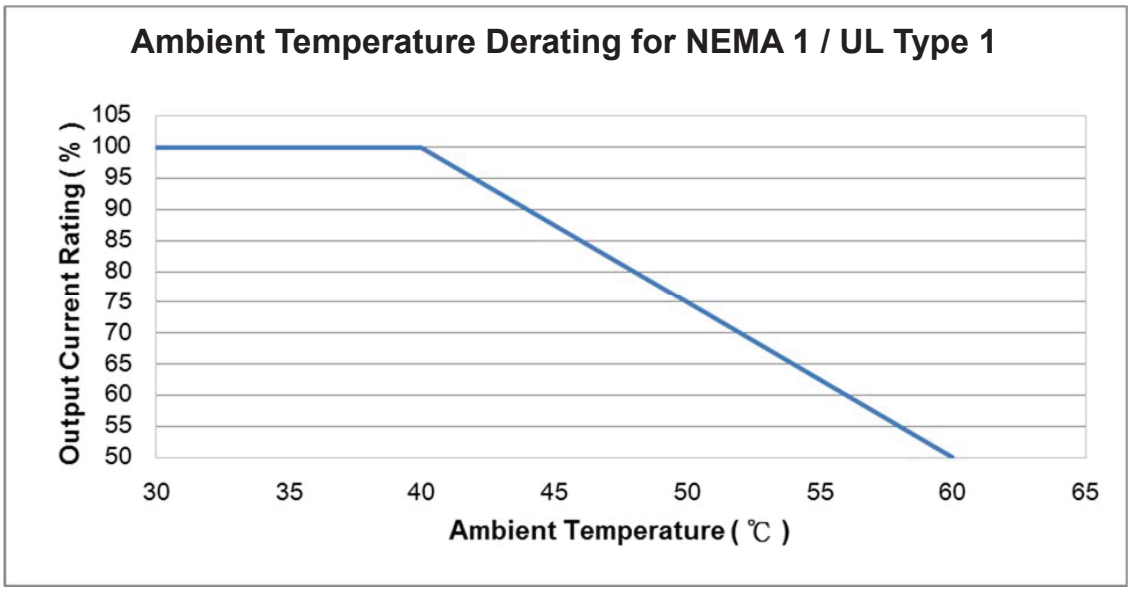
AC drives generate a significant amount of heat and will cause the internal temperature of an enclosure to exceed the rating of the AC drive, even when the ambient temperature is less than 104°F (40°C). Enclosure ventilation and/or cooling may be required to maintain a maximum internal temperature of 104°F (40°C) or less. Ambient temperature measurements/calculations should be made for the maximum expected temperature. When permissible, flange mounting the AC drive (mounting with the drive heatsink in open ambient air) can greatly reduce heating in the enclosure.

For use above 104˚F (40˚C), the AC drive must be derated as described below.

Derate Output Current Based on Temperature Above 104°F (40°C)

Drive Derating by Temperature and Protection Level	
Protection Level	***Derating***
UL Open Type / IP20 *	If the AC motor drive operates at the rated current, the ambient temperature needs to be between -20–50°C. If the temperature is above 50°C, decrease 2.5% of the rated current for every 1°C increase in temperature. The maximum allowable temperature is 60°C.
NEMA 1 / UL Type 1*	When the AC motor drive is operating at the rated current, the ambient temperature must be between -20–40°C. When the temperature is over 40 °C, for every increase by 1°C, decrease the rated current 2.5%. The maximum allowable temperature is 60°C.
** For more information about environmental ratings, refer to the "DURApulse GS10 AC Drive Environmental Information" on page 1–5 of this chapter.*	




Ambient Temperature Derating for NEMA 1 / UL Type 1
Output Current Rating (%)
105
100
95
90
85
80
75
70
65
60
55
50
30
35
40
45
50
55
60
65
Ambient Temperature (℃)

Derate Output Current Based on Carrier Frequency (if necessary)

Carrier Frequency Effects

AC Drives rectify the incoming 50 or 60Hz line power resulting in DC power at 0Hz. The resulting DC power is then pulse-width modulated and supplied to the motor by the drive's power electronics. IGBTs invert the DC power, simulating a sine wave at the desired frequency (that's what allows variable speed in AC induction motors). The speed at which the IGBTs are turned ON and OFF is called Carrier Frequency. In AC drives, the Carrier Frequency can range from 2kHz to 15kHz. The Carrier Frequency can be adjusted in most AC Drives.

There are trade-offs between choosing High Carrier Frequencies and Low Carrier Frequencies.

Benefits of Higher Carrier Frequencies:

- *Better efficiency (lower harmonic losses) in the motor*
- *Lower audible noise*

Benefits of Lower Carrier Frequencies:

- *Better efficiency in the drive*
- *Lower EMI (electrical noise)*
- *Reduced reflective wave peak voltage*

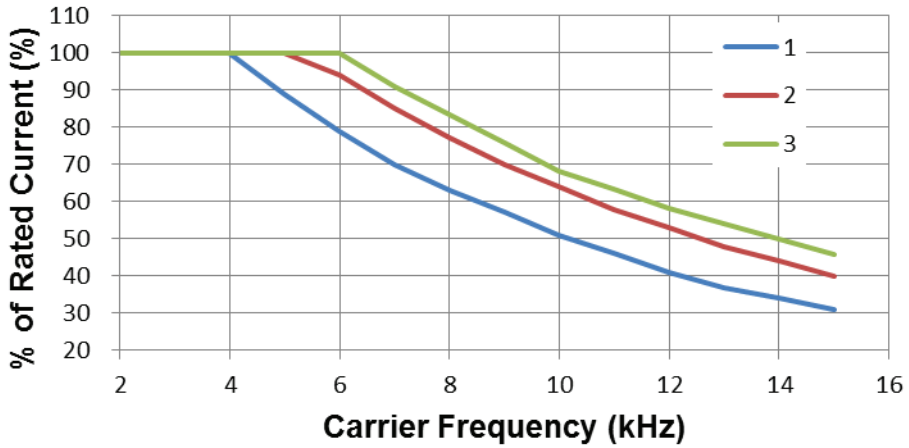
As a general rule, the Carrier Frequency should be set as low as possible without creating unacceptable audible noise in the motor. Smaller systems can have higher Carrier Frequencies, but larger drives (>20hp) should not have Carrier Frequencies set higher than 6kHz. Constant torque applications typically run around 2~4kHz.

GS10 Variable Torque Carrier Frequency Derating

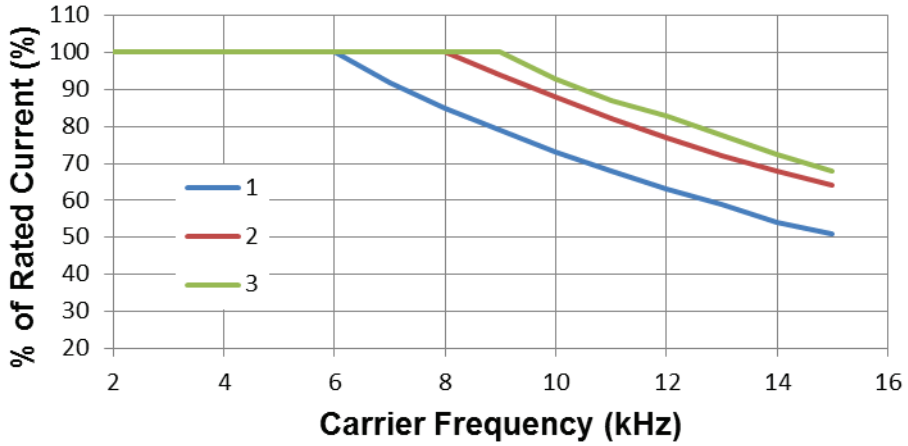
- *Line 1: Ta = 50°C / Load = 100%*
- *Line 2: Ta = 50°C / Load = 75% or Ta = 40°C / Load = 100%*
- *Line 3: Ta = 50°C / Load = 50% or Ta = 35°C / Load = 100%*

Note: Space Vector Pulse Width Modulation (SVPWM) and Two-Phase Pulse Width Modulation (DPWM) are determined by parameter P11.41. See Chapter 4 for details.

Variable Torque, SVPWM Mode

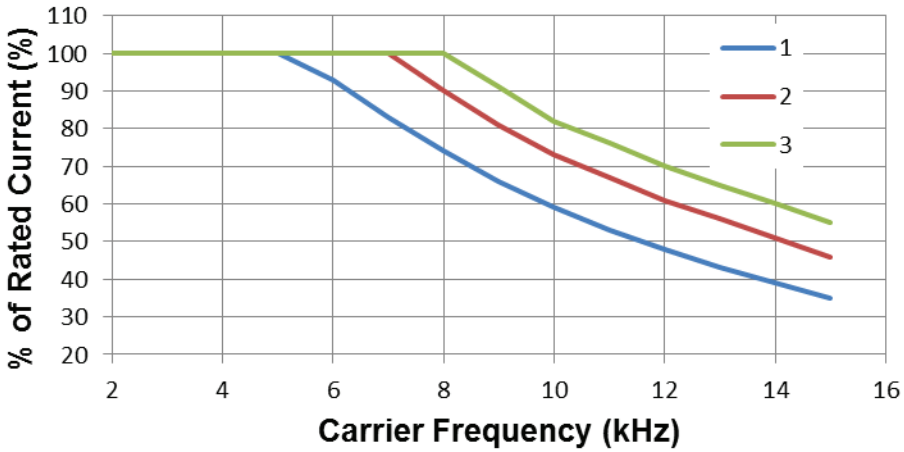


Variable Torque, DPWM Mode

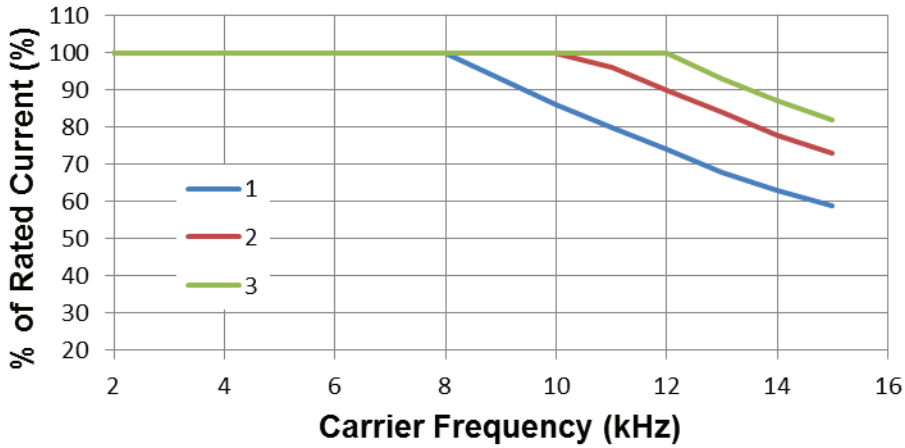


GS10 Constant Torque Carrier Frequency Derating

Constant Torque, SVPWM Mode



Constant Torque, DPWM Mode



DURAPULSE GS10 AC Drive Environmental Information

Storage and Transportation

AC drives should be kept in the shipping cartons or crates until they are installed. In order to retain the warranty coverage, they should be stored as described below if not to be installed and used within three months.

- *Store in a clean and dry location free from direct sunlight and corrosive fumes.*
- *Store within environmental conditions shown below in the "Environmental Conditions" table.*
- *DO NOT store in an area with rapid changes in temperature , to avoid condensation and frost.*
- *DO NOT place directly on the ground.*

If the drive is stored or is otherwise unused for more than a year, the drive's internal DC link capacitors should be recharged before use. Otherwise, the capacitors may be damaged when the drive starts to operate. We recommend recharging the capacitors of any unused drive at least once per year. (Refer to Chapter 6, "Maintenance and Troubleshooting" for information about recharging DC link capacitors.)

GS10 Environmental Conditions

Environmental Conditions for GS10 AC Drives			
Condition	Operation	Storage	Transportation
Installation Location	IEC 60364-1/ IEC 60664-1 Pollution degree 2, Indoor use only.	n/a	n/a
Ambient Temperature	IP20/UL Open Type: -20–50°C (-20–60°C w/derating)	-40–85°C	-20–70°C
	IP20 side-by-side or NEMA1/UL Type 1: -20–40°C (-20–55°C w/derating)		
	Non-condensing, non-freezing		
Relative Humidity	90%, no water condensation	95%, no water condensation	
Air Pressure	86–106 kPa	70–106 kPA	
Pollution Level	Concentrate prohibited		
	Class 3C2; Class 3S2	Class 2C2; Class 2S2	Class 1C2; Class 1S2
Altitude	<1000 m (For altitudes > 1000 m, derating required)		
Package Drop	n/a	ISTA procedure 1A (according to weight) IEC 60068-2-31	
Vibration	1.0 mm, peak to peak value range from 2–13.2 Hz; 0.7–1.0 G range from 13.2–55 Hz; 1.0 G range from 55–512 Hz. Compliance with IEC 60068-2-6	2.5 G peak, 5 Hz–2 kHz 0.015" maximum displacement	
Impact	15 G, 11 ms, compliance with IEC/EN60068-2-27	30G	
DO NOT expose the GS10 AC Drive to harsh environments such as dust, direct sunlight, corrosive/flammable gases, humidity, liquid, or vibrations. The salts in the air must be less than 0.01 mg/cm² every year.			

GS10 General Specifications

General Specifications for GS10 AC Drives			
Control Characteristics	Control Method	V/F, Sensorless Vector (SVC)	
	Applicable Motor	IM (Induction Motor), Permanent Magnet AC (IPM and SPM)	
	Starting Torque[1]	150% / 3 Hz 100% / (motor rated frequency/20)	(V/F, SVC control for IM, CT) (SVC control for PM, CT)
	Speed Control Range[1]	1: 50 (V/F, SVC control for IM, CT) 1: 20 (SVC control for PM, CT)	
	Max. Output Frequency	0.00–599.00 Hz	
	Overload Capacity	VT: rated output current of 120% 60 sec., 150% 3 sec. CT: rated output current of 150% 60 sec., 200% 3 sec.	
	Frequency Setting Signal	0–10 V / 4(0)–20 mA PWM pulse width input, pulse input (10kHz)	
	Digital Inputs	Five (5) - 24VDC NPN or PNP, includes 1 frequency input 10kHz	
	Digital Outputs	Two (2) - (1)-48VDC, (1) Relay-250VAC/30VDC	
	Analog Inputs	One (1) - Selectable Voltage or Current	
	Analog Outputs	One (1) - Voltage	
	Main Functions	• Multiple motor switching (max 2 motor settings) • Fast start-up • Deceleration Energy Back (DEB) function • Fast deceleration function • Master and Auxiliary frequency source selectable • Restart after momentary power loss • Speed tracking • Over-torque detection • 16-step speed (including the master speed) • Accel./decel. time switch • S-curve accel./decel. • Three-wire operation control • JOG frequency, • Frequency upper/lower limit settings • DC brake at start-up and stop • PID control • Simple Positioning Function • Multi Pump Sequence • RS-485 Serial Communications	
	Application Macro	Built-in application parameter groups (selected by industry) and user-defined application parameter groups.	
Protection Characteristics	Motor Protection	Over-current, Over-voltage, Over-heating, Phase loss, Over-load.	
	Stall Prevention	Stall prevention during acceleration, deceleration and running (independent settings).	
Agency Approvals	UL, CE, REACH		
1: Control accuracy may vary depending on the environment, application conditions, or motor type. For more information contact AutomationDirect.			

Efficiency Class

The EU Ecodesign regulation directive establishes a framework to set mandatory ecological requirements for energy-using and energy-related products. The IEC 61800-9-2 standard defines the efficiency classes for AC drives. The efficiency classes range (low to high) from IE0 to IE2. These classes apply to AC drives rated 100 to 1000 V and 0.12 to 1000 kW (1/6 to 1,340 HP).

Drive manufacturers must declare power losses in terms of percentage of rated apparent output power at eight different operating points, as well as standby losses. The International Efficiency (IE) level is given at the nominal point.

The power losses of GS10 series drives shall not exceed the maximum power losses corresponding to the IE2 efficiency level. For specific power losses of each drive model, see the drive specification tables.

DURAPULSE GS10 AC DRIVE SPECIFICATIONS

120V CLASS – 1-PHASE MODEL-SPECIFIC SPECIFICATIONS

GS10 120V Class Specifications; Frame Size A, C [1]						
Model Name: GS11N-1xPx				**GS11N-10P2**	**GS11N-10P5**	**GS11N-11P0**
Frame Size				**A1**	**A3**	**C1**
Output Rating	**Max Motor Output**		**hp**	1/4	1/2	1
			kW	0.2	0.4	0.75
	CT	**Rated Output Capacity**	**kVA**	0.6	1.0	1.8
		Rated Output Current	**A**	1.6	2.5	4.8
		Carrier Frequency[4]	**kHz**	2–15		
	VT	**Rated Output Capacity**	**kVA**	0.7	1.0	2.1
		Rated Output Current	**A**	1.8	2.7	5.5
		Carrier Frequency[4]	**kHz**	2–15		
Input Rating[2]	**CT**	**Rated Input Current**	**A**	6	9.4	18
	VT	**Rated Input Current**	**A**	6.8	10.1	20.6
	Rated Voltage/Frequency[3]			One-phase: 100–120 VAC (-15% to +10%), 50/60 Hz		
	Operating Voltage Range (VAC)			85–132		
	Frequency Tolerance (Hz)			47–63		
IE2 Efficiency - Relative Power Loss				4.3%	3.2%	2.9%
Weight (kg)				0.4	0.5	1
Cooling Method				Convective		Fan
IP Rating				IP20		

1 - For use with three-phase motors only.
2 - Please refer to Appendix A - Accessories for input fusing information.
3 - For 120V single phase input power, remove the drive RFI jumper. See Circuit Connections - RFI Jumpers in Chapter 2.
4 - If application requires adjustment of the carrier frequency above default, refer to "Derate Output Current Based on Carrier Frequency (if necessary)" on page 1–7

230V Class – 1-Phase Model-Specific Specifications

GS10 230V Class Specifications; Frame Size A, B, C[1]								
Model Name: GS11N-2xPx				**GS11N-20P2**	**GS11N-20P5**	**GS11N-21P0**	**GS11N-22P0**	**GS11N-23P0**
Frame Size				**A1**	**A3**	**B2**	**C1**	**C1**
Output Rating	**Max Motor Output**		**hp**	1/4	1/2	1	2	3
			kW	0.2	0.4	0.75	1.5	2.2
	CT	**Rated Output Capacity**	**kVA**	0.6	1.1	1.8	2.9	4.2
		Rated Output Current	**A**	1.6	2.8	4.8	7.5	11
		Carrier Frequency[4]	**kHz**	2–15				
	VT	**Rated Output Capacity**	**kVA**	0.7	1.2	1.9	3.2	4.8
		Rated Output Current	**A**	1.8	3.2	5	8.5	12.5
		Carrier Frequency[4]	**kHz**	2–15				
Input Rating[2]	**CT**	**Rated Input Current**	**A**	5.1	7.3	10.8	16.5	24.2
	VT	**Rated Input Current**	**A**	5.8	8.3	11.3	18.5	27.5
	Rated Voltage/Frequency[3]			One-phase 200-240 VAC (-15% to +10%), 50/60 Hz				
	Operating Voltage Range (VAC)			170–265				
	Frequency Tolerance (Hz)			47–63				
IE2 Efficiency - Relative Power Loss				4.7%	3.1%	2.7%	2.5%	2.4%
Weight (kg)				0.4	0.5	0.8	1	1
Cooling Method				Convective				Fan
IP Rating				IP20				

1 - For use with three-phase motors only.
2 - Please refer to "Appendix A - Accessories" for input fusing information.
3 - For input power that is a floating ground or IT type, the RFI jumper must be removed. See "Floating Ground System (IT Systems)" on page 2–12.
4 - If application requires adjustment of the carrier frequency above default, refer to "Derate Output Current Based on Carrier Frequency (if necessary)" on page 1–7.

230V Class – 3-Phase Model-Specific Specifications

GS10 230V Class Specifications; Frame Size A, B[1]							
Model Name: GS13N-2xPx				**GS13N-20P2**	**GS13N-20P5**	**GS13N-21P0**	**GS13N-22P0**
Frame Size				**A1**	**A2**	**A5**	**B1**
Output Rating	**Max Motor Output (3-phase [1-phase])**		**hp**	1/4 [0.1]	1/2 [1/4]	1 [1/2]	2 [1]
			kW	0.2 [0.1]	0.4 [0.2]	0.75 [0.375]	1.5 [0.75]
	CT	**Rated Output Capacity**	**kVA**	0.6	1.1	1.8	2.9
		Rated Output Current (3-phase [1-phase])	**A**	1.6 [0.8]	2.8 [1.4]	4.8 [2.4]	7.5 [3.75]
		Carrier Frequency[4]	**kHz**	2–15			
	VT	**Rated Output Capacity**	**kVA**	0.7	1.2	1.9	3.0
		Rated Output Current	**A**	1.8	3.0	5.0	8.0
		Carrier Frequency[4]	**kHz**	2–15			
Input Rating[2]	**CT**	**Rated Input Current (3-phase and 1-phase)**	**A**	1.9	3.4	5.8	9.0
	VT	**Rated Input Current**	**A**	2.2	3.8	6.0	9.6
	Rated Voltage/Frequency[3]			3-phase 200-240 VAC (-15% to +10%), 50/60 Hz			
	Operating Voltage Range (VAC)			170–265			
	Frequency Tolerance (Hz)			47–63			
IE2 Efficiency - Relative Power Loss				4.7%	3.1%	2.7%	2.4%
Weight (kg)				0.4	0.5	0.6	0.8
Cooling Method				Convective			Fan
IP Rating				IP20			

1 - For use with three-phase motors only.
2 - Please refer to "Appendix A - Accessories" for input fusing information.
3 - For input power that is a floating ground or IT type, the RFI jumper must be removed. See "Floating Ground System (IT Systems)" on page 2–12.
4 - If application requires adjustment of the carrier frequency above default, refer to "Derate Output Current Based on Carrier Frequency (if necessary)" on page 1–7.

230V Class – 3-Phase Model-Specific Specifications

GS10 230V Class Specifications; Frame Size C, D[1]						
Model Name: GS13N-2xPx				**GS13N-23P0**	**GS13N-25P0**	**GS13N-27P5**
Frame Size				**C1**	**C1**	**D1**
Output Rating	**Max Motor Output (3-phase [1-phase])**		**hp**	3 [1.5]	5 [2.5]	7.5 [3.5]
			kW	2.2 [1.1]	3.7 [1.85]	5.5 [2.75]
	CT	**Rated Output Capacity**	**kVA**	4.2	6.5	9.5
		Rated Output Current (3-phase [1-phase])	**A**	11 [5.5]	17 [8.5]	25 [12.5]
		Carrier Frequency[4]	**kHz**	2–15		
	VT	**Rated Output Capacity**	**kVA**	4.8	7.4	10.3
		Rated Output Current	**A**	12.5	19.5	27
		Carrier Frequency[4]	**kHz**	2–15		
Input Rating[2]	**CT**	**Rated Input Current (3-phase and 1-phase)**	**A**	13.2	20.0	30.0
	VT	**Rated Input Current**	**A**	15.0	23.4	32.4
	Rated Voltage/Frequency[3]			3-phase 200-240 VAC (-15% to +10%), 50/60 Hz		
	Operating Voltage Range (VAC)			170–265		
	Frequency Tolerance (Hz)			47–63		
IE2 Efficiency - Relative Power Loss				2.4%	2.2%	2.3%
Weight (kg)				1	1	2
Cooling Method				Fan		
IP Rating				IP20		

1 - For use with three-phase motors only.
2 - Please refer to "Appendix A - Accessories" for input fusing information.
3 - For input power that is a floating ground or IT type, the RFI jumper must be removed. See "Floating Ground System (IT Systems)" on page 2–12.
4 - If application requires adjustment of the carrier frequency above default, refer to "Derate Output Current Based on Carrier Frequency (if necessary)" on page 1–7.

460V Class – 3-Phase Model-Specific Specifications

GS10 460V Class Specifications; Frame Size A, B[1]						
Model Name: GS13N-4xPx				**GS13N-40P5**	**GS13N-41P0**	**GS13N-42P0**
Frame Size				**A4**	**A6**	**B1**
Output Rating	**Max Motor Output**		**hp**	1/2	1	2
			kW	0.4	0.75	1.5
	CT	**Rated Output Capacity**	**kVA**	1.1	2.1	3.2
		Rated Output Current	**A**	1.5	2.7	4.2
		Carrier Frequency[4]	**kHz**	2–15		
	VT	**Rated Output Capacity**	**kVA**	1.4	2.3	3.5
		Rated Output Current	**A**	1.8	3.0	4.6
		Carrier Frequency[4]	**kHz**	2–15		
Input Rating[2]	**CT**	**Rated Input Current**	**A**	2.1	3.7	5.8
	VT	**Rated Input Current**	**A**	2.5	4.2	6.4
	Rated Voltage/Frequency[3]			3-phase 380-480 VAC (-15% to +10%), 50/60 Hz		
	Operating Voltage Range (VAC)			323-528		
	Frequency Tolerance (Hz)			47–63		
IE2 Efficiency - Relative Power Loss				3.7%	2.5%	2.2%
Weight (kg)				0.6	0.7	0.8
Cooling Method				Convective		Fan
IP Rating				IP20		

1 - For use with three-phase motors only.
2 - Please refer to "Appendix A - Accessories" for input fusing information.
3 - For input power that is a floating ground or IT type, the RFI jumper must be removed. See "Floating Ground System (IT Systems)" on page 2–12.
4 - If application requires adjustment of the carrier frequency above default, refer to "Derate Output Current Based on Carrier Frequency (if necessary)" on page 1–7.

460V Class – 3-Phase Model-Specific Specifications

GS10 460V Class Specifications; Frame Size C, D[1]						
Model Name: GS13N-4xPx				**GS13N-43P0**	**GS13N-45P0**	**GS13N-47P5**	**GS13N-4010**
Frame Size				**C1**	**C1**	**D1**	**D1**
Output Rating	**Max Motor Output**		**hp**	3	5	7 1/2	10
			kW	2.2	3.7	5.5	7.5
	CT	**Rated Output Capacity**	**kVA**	4.2	6.9	9.9	13
		Rated Output Current	**A**	5.5	9	13	17.5
		Carrier Frequency[4]	**kHz**	2–15			
	VT	**Rated Output Capacity**	**kVA**	5.0	8.0	12	15.6
		Rated Output Current	**A**	6.5	10.5	14.5	19.8
		Carrier Frequency[4]	**kHz**	2–15			
Input Rating[2]	**CT**	**Rated Input Current**	**A**	6.1	9.9	14.3	19.3
	VT	**Rated Input Current**	**A**	7.2	11.6	16.0	21.8
	Rated Voltage/Frequency[3]			3-phase 380-480 VAC (-15% to +10%), 50/60 Hz			
	Operating Voltage Range (VAC)			323-528			
	Frequency Tolerance (Hz)			47–63			
IE2 Efficiency - Relative Power Loss				2.3%	2.0%	1.9%	1.9%
Weight (kg)				1	1	2	2
Cooling Method				Fan			
IP Rating				IP20			

1 - For use with three-phase motors only.
2 - Please refer to "Appendix A - Accessories" for input fusing information.
3 - For input power that is a floating ground or IT type, the RFI jumper must be removed. See "Floating Ground System (IT Systems)" on page 2–12.
4 - If application requires adjustment of the carrier frequency above default, refer to "Derate Output Current Based on Carrier Frequency (if necessary)" on page 1–7.

NOTE: 120VAC models do not have DC bus terminals.

Receiving and Inspection

Drive Package Contents

After receiving the GS10 AC drive, please check the following:

1) Make sure that the package includes the DURAPULSE GS10 AC drive and the Quick-Start Guide that matches your product.
2) Please inspect the unit after unpacking to assure it was not damaged during shipment. Make sure that the part number printed on the package corresponds with the part number indicated on the nameplate.
3) Make sure that the part number indicated on the nameplate corresponds with the part number of your order.
4) Make sure that the voltage for the wiring lies within the range as indicated on the nameplate. Please install the GS10 AC drive according to this manual.
5) Before applying the power, please make sure that all the devices, including power, motor, control board, and digital keypad are connected correctly.
6) When wiring the GS10 AC drive, please make sure that the wiring of input terminals “R/L1, S/L2, T/L3” and output terminals “U/T1, V/T2, W/T3” are correct to prevent drive damage.
7) When power is applied, select the language and set parameter groups via the digital keypad. When executing a trial run, please begin with a low speed, and then gradually increase the speed until the desired speed is reached.

The GS10 AC drive should be kept in the shipping carton before installation. In order to retain the warranty coverage, the GS10 AC drive should be stored properly when it is not to be used for an extended period of time. Refer to the preceding “Environmental Information” section for proper storage conditions.

Model Number Explanation

GS11N- 4 7P5

Applicable Motor Capacity*

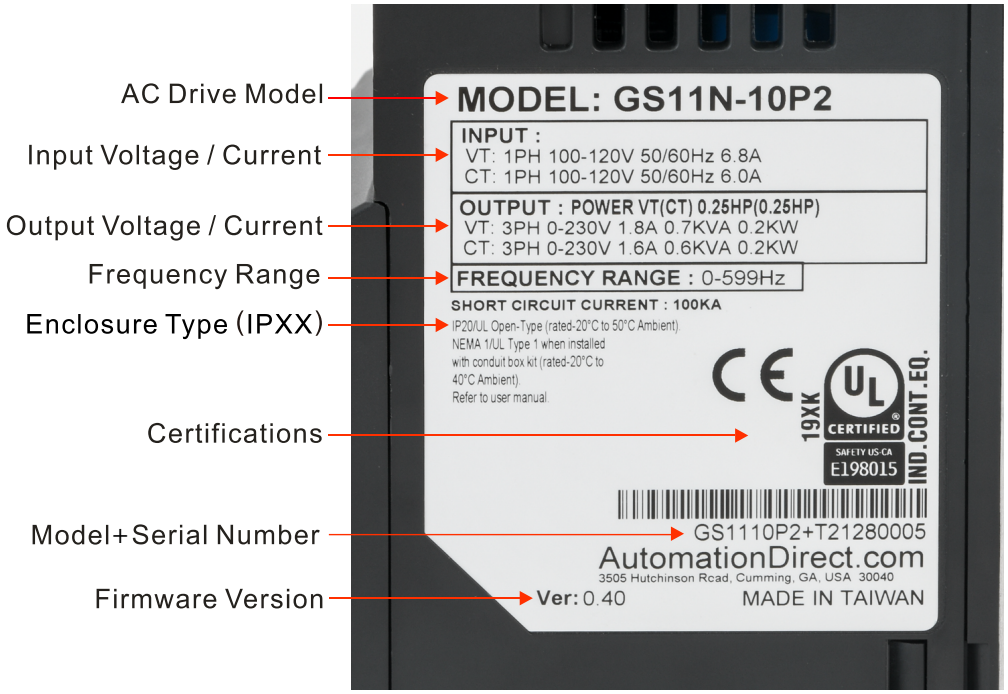
0P2: 0.25hp	0P5: 0.5hp	1P0: 1.0hp	2P0: 2.0hp
3P0: 3.0hp	5P0: 5.0hp	7P5: 7.5hp	010: 10hp

*Not all capacities are available in each voltage.

Input Voltage
1: 120VAC
2: 230VAC
4: 460VAC

Model Line
N: No Safe Torque Off (STO)
A: Accessories

Phase
0: Not phase-specific
1: One-phase
3: Three-phase

Series Name

Nameplate Information